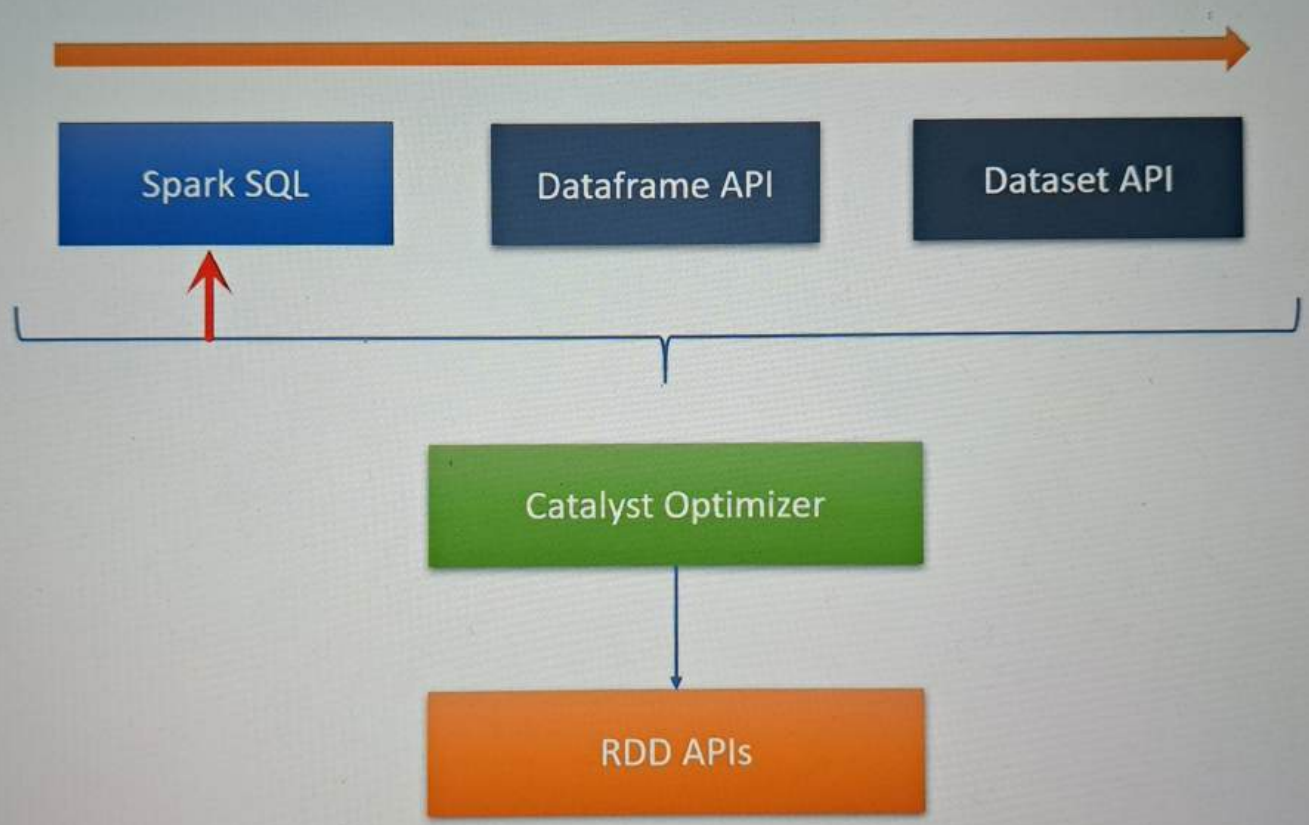
Spark APIs



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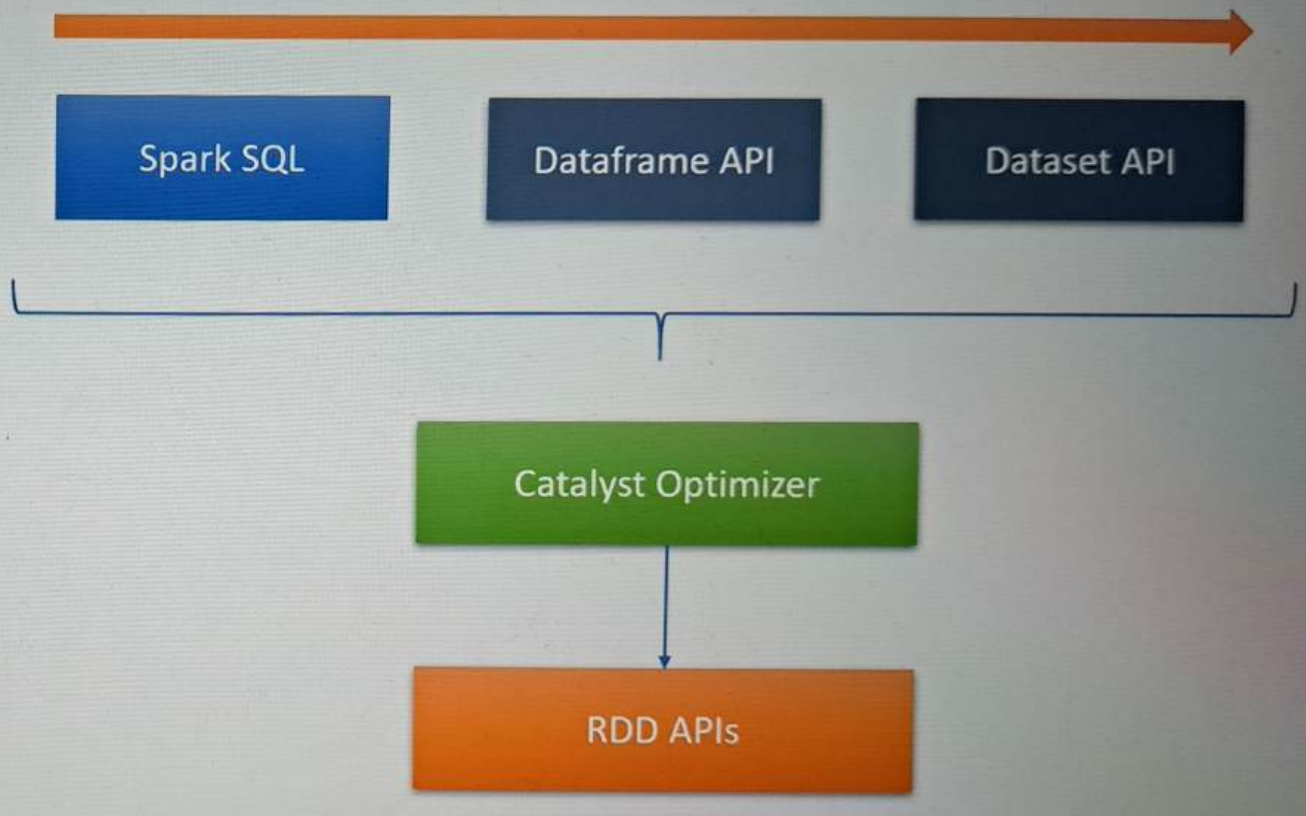
Spark APIs

In this section ...

1. Spark RDD API Example
2. Spark SQL Example
3. SQL Engine & Catalyst Optimizer



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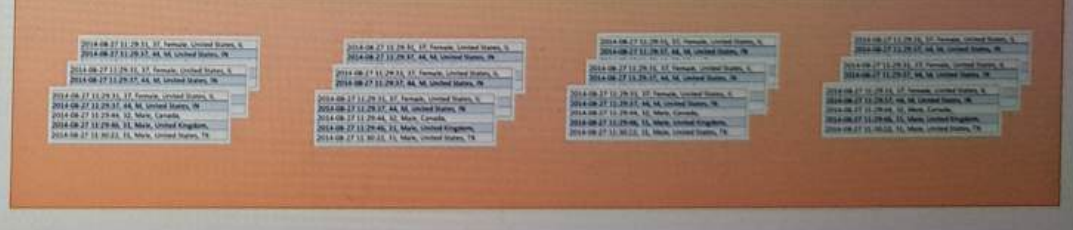
RDD – Resilient Distributed Dataset

Dataset –

2014-08-27 11:29:31, 37, Female, United States, IL
2014-08-27 11:29:37, 44, M, United States, IN
2014-08-27 11:29:44, 32, Male, Canada,
2014-08-27 11:29:46, 31, Male, United Kingdom,
2014-08-27 11:30:22, 31, Male, United States, TX

Distributed –



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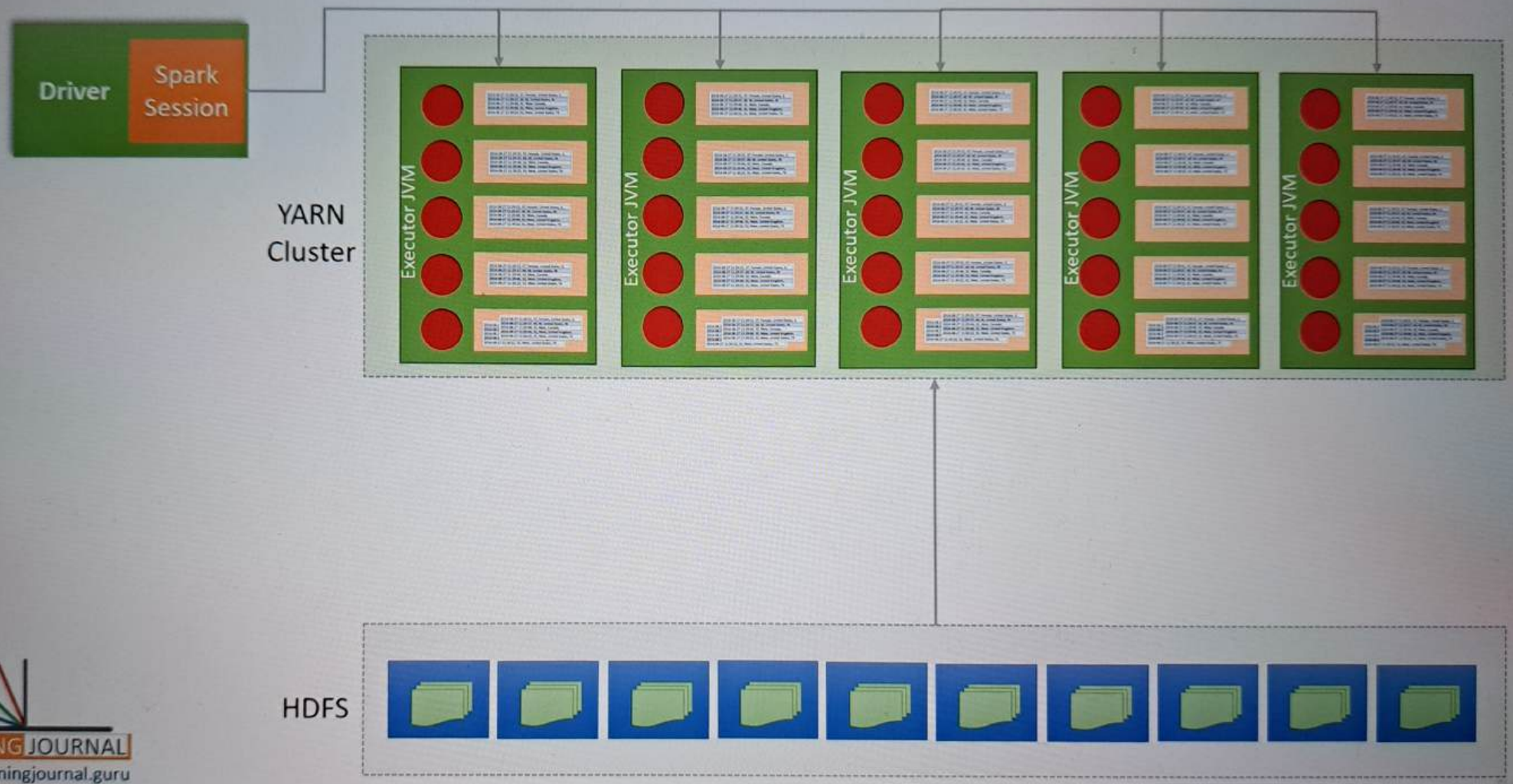
Spark RDD APIs



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How to Create RDD?

ûdem

```
14      spark = SparkSession \
15          .builder \
16          .config(conf=conf) \
17          .getOrCreate()
18
19      sc = spark.sparkContext
20      logger = Log4j(spark)
21
22      if len(sys.argv) != 2:
23          logger.error("Usage: HelloSpark <filename>")
24          sys.exit(-1)
25
26      linesRDD = sc.textFile(sys.argv[1])
27
```

How to process RDD?

Udemy

```python
    logger = Log4j(spark)

    if len(sys.argv) != 2:
        logger.error("Usage: HelloSpark <filename>")
        sys.exit(-1)

    linesRDD = sc.textFile(sys.argv[1])
    partitionedRDD = linesRDD.repartition(2)

    colsRDD = partitionedRDD.map(lambda line: line.replace('"', '').split(","))
    selectRDD = colsRDD.map(lambda cols: SurveyRecord(int(cols[1]), cols[2], cols[3], cols[4]))
    filteredRDD = selectRDD.filter(lambda r: r.Age < 40)
    kvRDD = filteredRDD.map(lambda r: (r.Country, 1))
    countRDD = kvRDD.reduceByKey(lambda v1, v2: v1 + v2)

    colsList = countRDD.collect()
    for x in colsList:
        logger.info(x)
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 03-HelloSparkSQL - HelloSparkSQL.py - PyCharm

03-HelloSparkSQL > HelloSparkSQL.py

Project

- 03-HelloSparkSQL C:\Users\prashant\Documents\GitHub\Spark
  - app-logs
  - data
  - lib
  - spark-warehouse
  - HelloSparkSQL.py
  - log4j.properties
- External Libraries
- Scratches and Consoles

HelloSparkSQL.py

```python
import sys
from pyspark.sql import SparkSession
from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("HelloSparkSQL") \
        .getOrCreate()

    logger = Log4j(spark)

    if len(sys.argv) != 2:
        logger.error("Usage: HelloSpark <filename>")
        sys.exit(-1)

    surveyDF = spark.read \
        .option("header", "true") \
        .option("inferSchema", "true") \
        .csv(sys.argv[1])

    surveyDF.createOrReplaceTempView("survey_tbl")
    countDF = spark.sql("select Country, count(1) as count from survey_tbl where Age<40 group by Country")
    countDF.show()
```

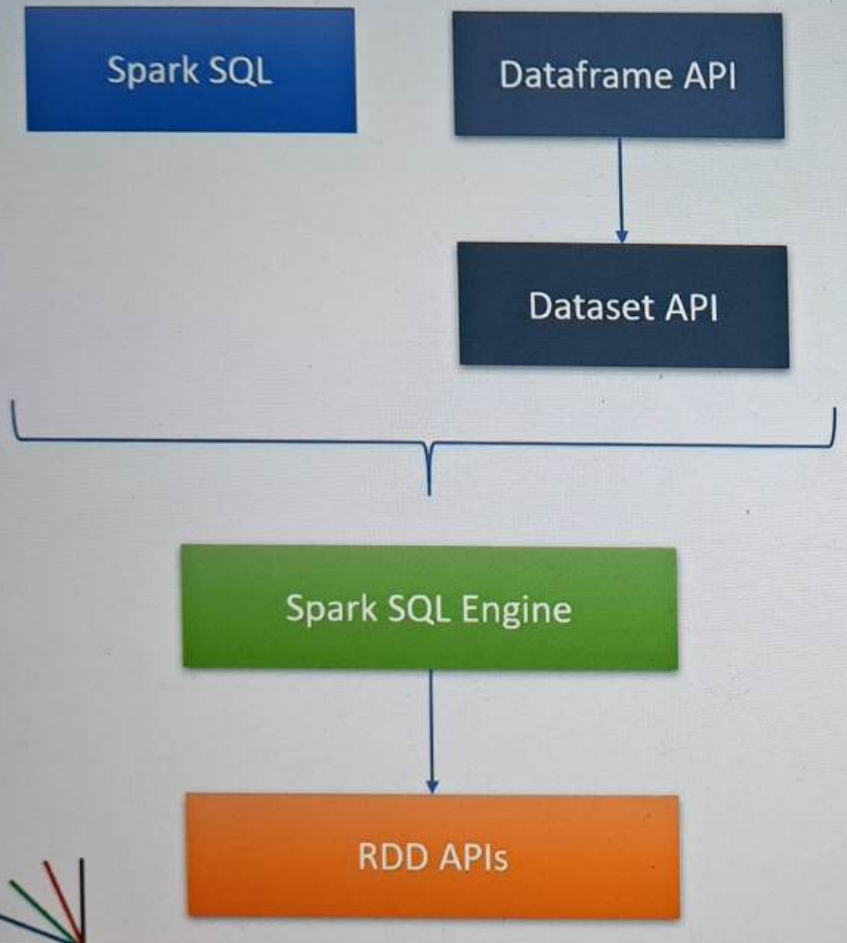
if __name__ == "__main__"

Git TODO Run Terminal Python Console

Event Log

19:36 CRLF UTF-8 4 spaces Python 3.7 (03-HelloSparkSQL) master

Spark APIs



1. Analysis
2. Logical Optimization
3. Physical Planning
4. Code Generation

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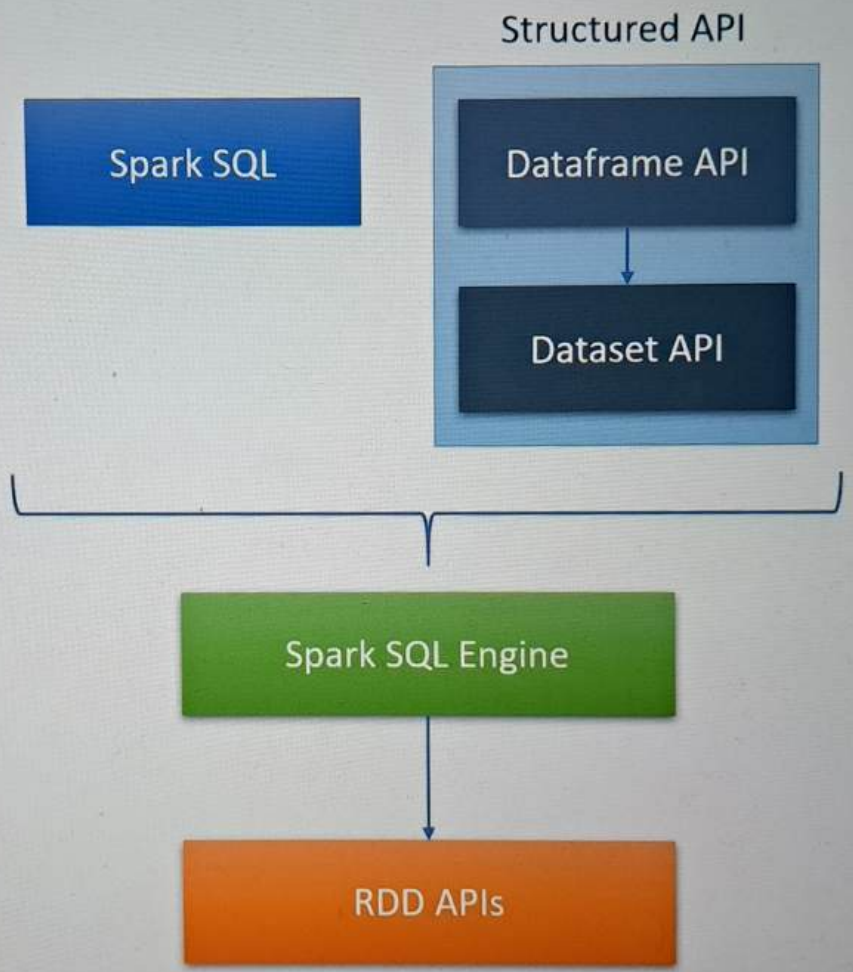
Spark APIs



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